ENGLISH PAPER SUMMARY | ALL ORIGINAL NUMBERED FIGURES

Modeling the ice-attenuated waves in the Great Lakes

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Adding ice attenuation and open-water source scaling suppresses unrealistically large winter waves, but the February 2011 retreat event exposes the importance of accurate ice evolution and missing wave-to-ice feedback.

Prepared independently from the original paper. This is a new English synthesis, not a translation of the Korean notes. The figure atlas reproduces all 7 numbered figures with explanatory reading notes. Section and equation references refer to the source publication.

Question, contribution, and coupling direction

The study develops a partly coupled Great Lakes wave-ice model within FVCOM. It asks how ice changes significant wave height, how sensitive that change is to seven empirical attenuation schemes, and whether under-ice wave observations support the approach. FVCOM-SWAVE is based on SWAN; UG-CICE supplies lake ice dynamics. WAVEWATCH III IC4 parameterizations are transplanted into SWAVE: the experiment is not a native WW3 simulation. [Sections 1-2.1]

Ice affects waves through blocking-related changes to source terms and an additional attenuation sink. Direct wave-induced ice breakup is absent. This is the central one-way limitation, even though hydrodynamics, ice, and waves are part of a larger coupled modeling framework.

Equations and experimental setup

The spectral wave action balance advances action density in physical, frequency, and directional space. In the ice-covered formulation, wind input, whitecapping, and breaking terms are scaled by the open-water fraction (1 - C_ice). Nonlinear wave-wave transfer and bottom friction retain their open-water form, and an ice sink S_ice = -alpha E C_g is added. Blocking of local generation and attenuation of propagating energy are distinct operations. [Equations 1-2]

The seven IC4 options range from wave-period relationships and frequency steps to relationships involving ice thickness or significant wave height. Their attenuation coefficients can differ by orders of magnitude. IC4M3 and IC4M7 depend on thickness; IC4M4 depends on wave height rather than frequency. Figure 2 and Table 1 show why parameterization choice matters. [Section 2.2]

The five-lake grid has approximately 3.5 km average horizontal resolution and 21 vertical layers. Atmospheric forcing uses 3-hourly, approximately 32 km NARR fields. The wave module uses 40 frequency bins from 0.04 to 1 Hz and 36 directions. The 2014 experiments compare a wave-only reference with seven ice-attenuated runs; the separate 2011 comparison uses IC4M6. [Section 2.3 and Table 2]

Validation design and main results

Eight NDBC buoys assess ice-free wave performance in 2014. Satellite-based NIC ice concentration assesses the ice field, and bottom-mounted AWAC wave observations at three Lake Erie stations assess winter behavior in February 2011. These tests answer different questions: an ice-free wave benchmark does not by itself validate ice attenuation. [Section 2.4]

Across the eight ice-free comparisons, the reported overall correlation is 0.77, RMSE 0.38 m, and MAE 0.26 m. NARR wind forcing generally underestimates observed winds and contributes to wave errors. These statistics describe the background wave model, not winter surrogate accuracy. [Section 3.1; Figures 3-4]

For February 2014, modeled ice and NIC data show extensive ice in Superior, Huron, and Erie, with more open water in central Michigan and Ontario. Across IC4M1-M7, mean significant-wave-height reductions over the Great Lakes excluding Ontario range from 0.324 to 0.467 m. Despite different magnitudes, spatial patterns are similar: high ice concentration substantially suppresses waves. The absence of simultaneous under-ice wave data in 2014 limits direct validation of these attenuation estimates. [Sections 3.2-3.3; Figures 5-6]

The February 2011 event is the critical diagnostic

During the high-ice intervals, the ice-aware run reproduces the near-zero observed waves substantially better than the wave-only run. During 18-20 February, however, ice observations decrease sharply and observed waves exceed 2 m. The ice-aware run retains too much ice and underpredicts the wave event, while the wave-only run is closer for that interval. [Section 4; Figure 7]

Table 3 makes this regime dependence explicit. At station 4a, the high-ice first interval has RMSE 0.05 m for the ice-aware experiment versus 0.54 m for the wave-only case. During the retreat interval, the corresponding RMSEs are 1.32 m and 0.47 m. A whole-season average would conceal this reversal in skill.

The authors discuss forcing errors, grid resolution, and omitted wave-induced ice breakage as possible contributors. Their analysis suggests an ice-retreat/wave-growth feedback but does not establish that breakup alone explains the error. Accurate and rapidly responding ice fields are a prerequisite for useful wave attenuation.

Limitations and implications for the research project

This study provides a practical numerical baseline and a clear example of conditional skill. Its empirical attenuation coefficients are not universal freshwater laws, its 2014 winter attenuation maps are not directly validated by winter wave observations, and its 2011 comparison does not demonstrate a fully coupled fracture process. [Sections 4-5]

For a Great Lakes surrogate, distinguish learning the attenuation coefficient or spectral sink from learning the wave model's final height. Keep errors caused by incorrect ice forcing separate from errors in the attenuation mapping. Evaluate freeze-up, stable ice, and retreat events separately, including the February 2011 failure case.

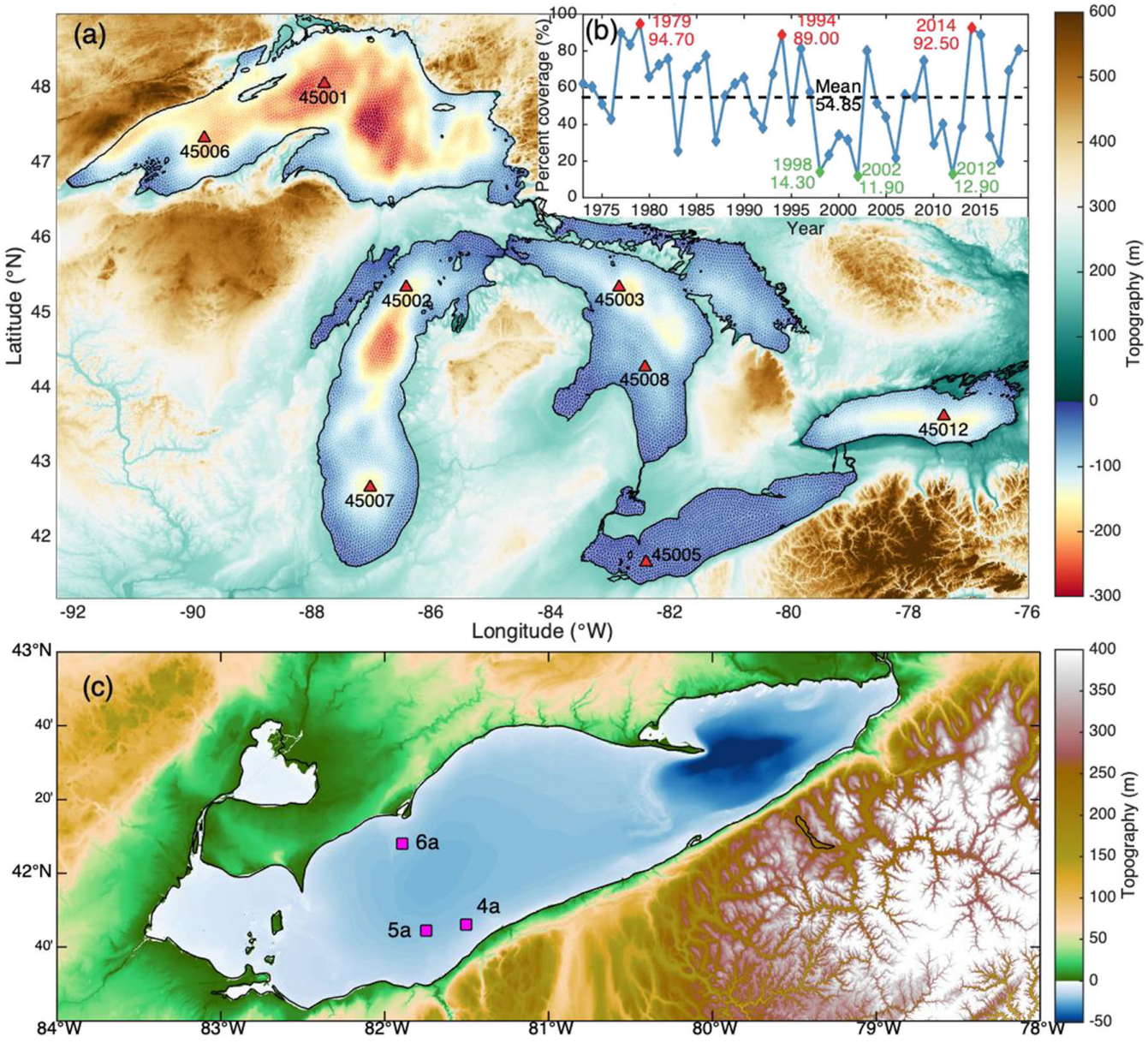
The 40-bin wave spectrum used here is not automatically interchangeable with the 25 spectral inputs used by WIFF1.0. Frequency bounds, bin-integrated energy, directional integration, and units require an explicit conversion. These are proposed project requirements, not procedures implemented in this paper.

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FIGURE ATLAS | 01 / 07

Figure 1. Great Lakes domain, historical ice, and moorings



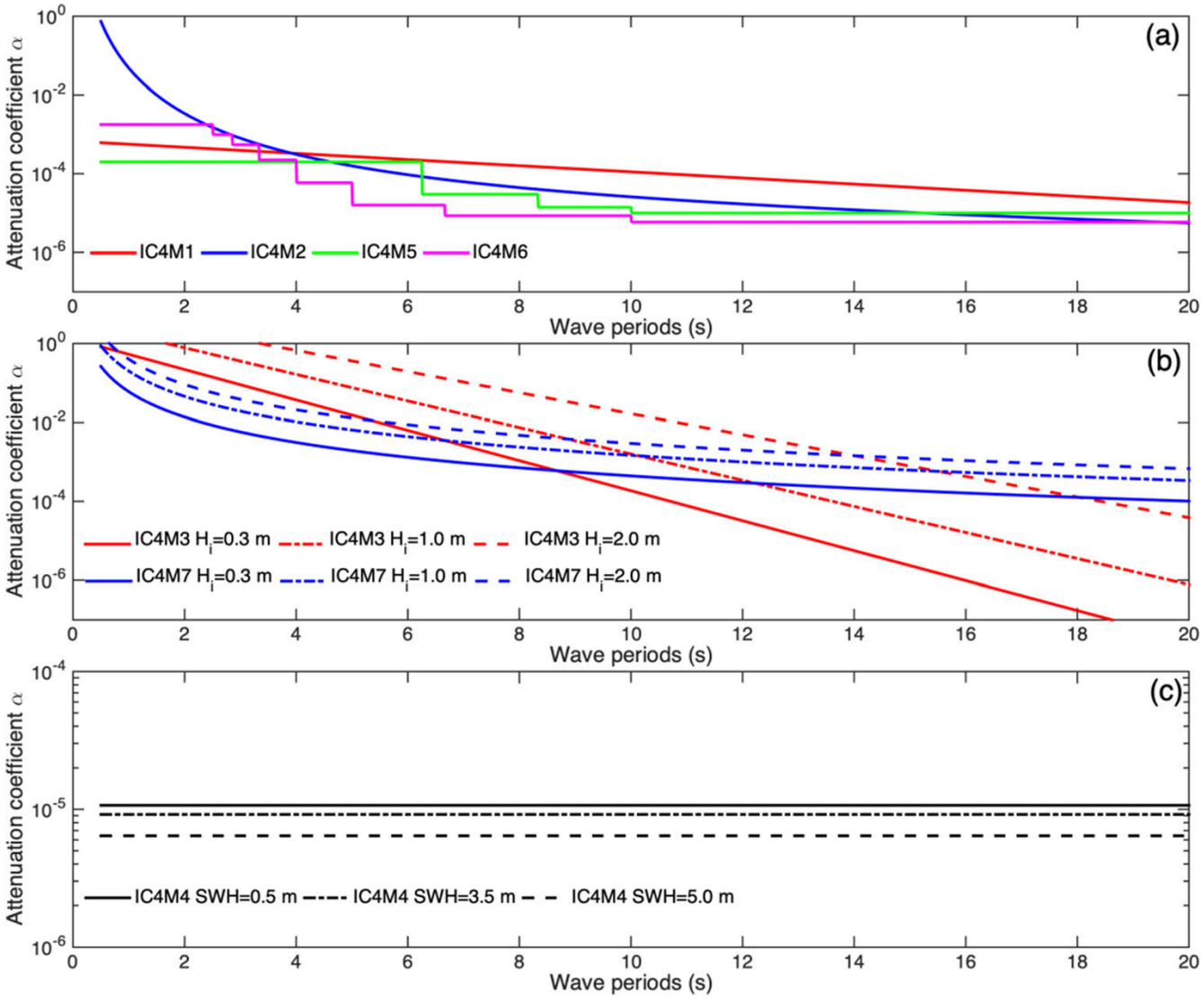
How to read this figure

The figure locates the unstructured model mesh, wave-validation buoys, historical maximum ice coverage, and Lake Erie AWAC sites. It separates the basin-wide spatial problem from the limited winter observational footprint. The mooring locations are essential for interpreting which regional claims have direct under-ice validation.

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FIGURE ATLAS | 02 / 07

Figure 2. Attenuation schemes are structurally different



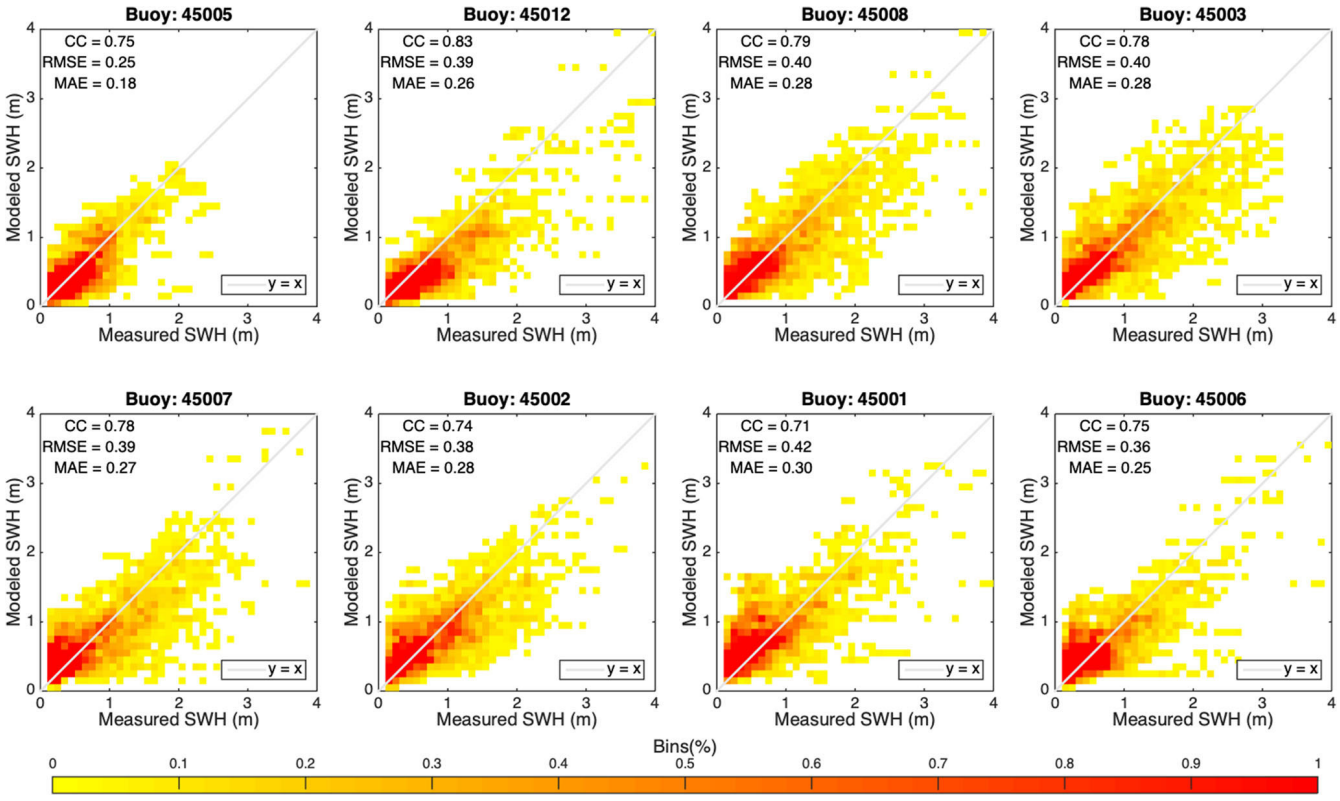
How to read this figure

The panels compare attenuation against wave period, including thickness-dependent and wave-height-dependent cases. Large differences are a consequence of both functional form and the conditions of the source datasets. Treating IC4M1-M7 as interchangeable labels would hide this physical and numerical diversity.

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FIGURE ATLAS | 03 / 07

Figure 3. Ice-free wave-model validation



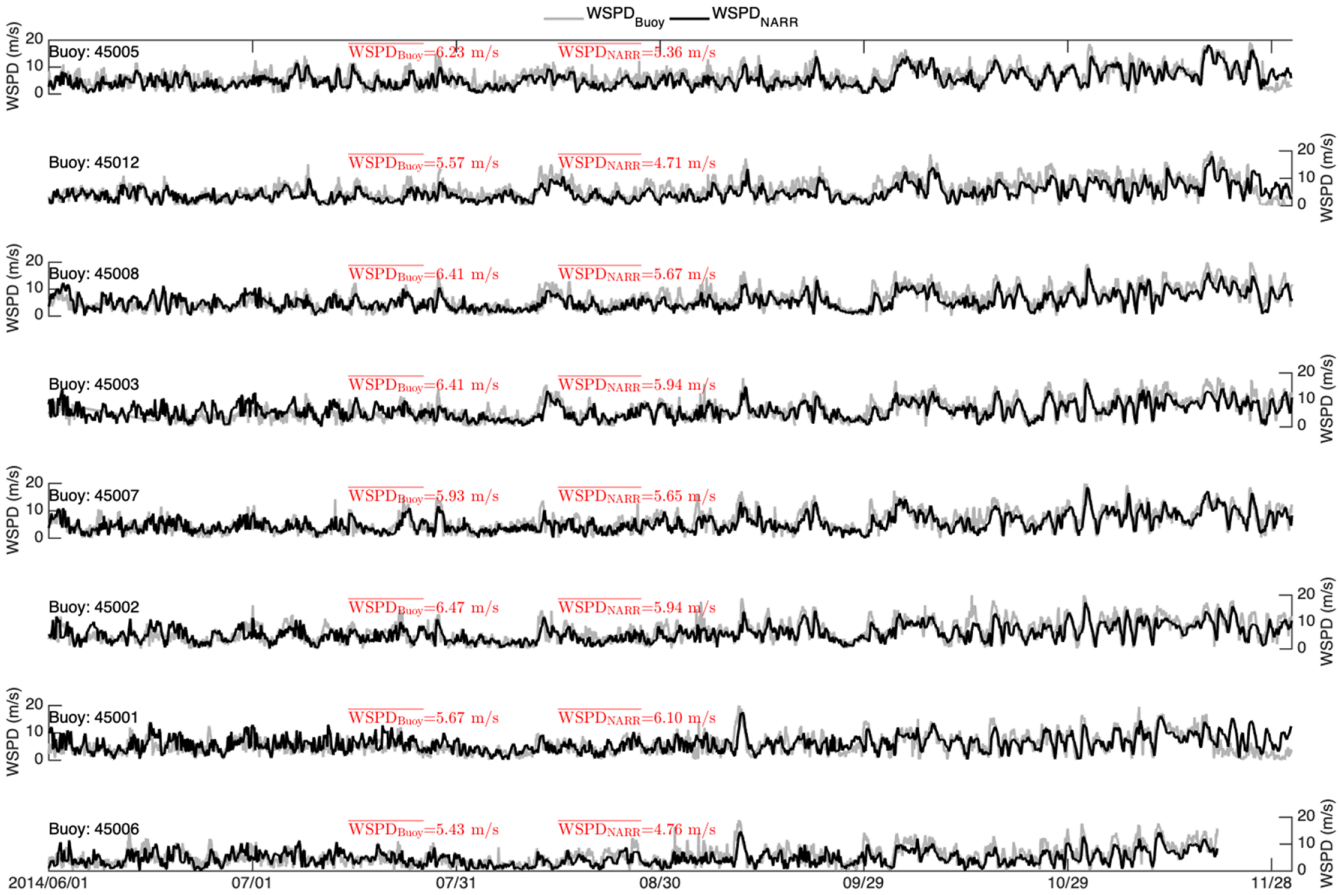
How to read this figure

Eight buoy comparisons assess modeled versus measured significant wave height in the 2014 ice-free season. Binned scatter and the agreement line reveal background wave skill. This figure supports the base wave model, but does not test wave attenuation inside ice.

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FIGURE ATLAS | 04 / 07

Figure 4. Wind forcing as an error source



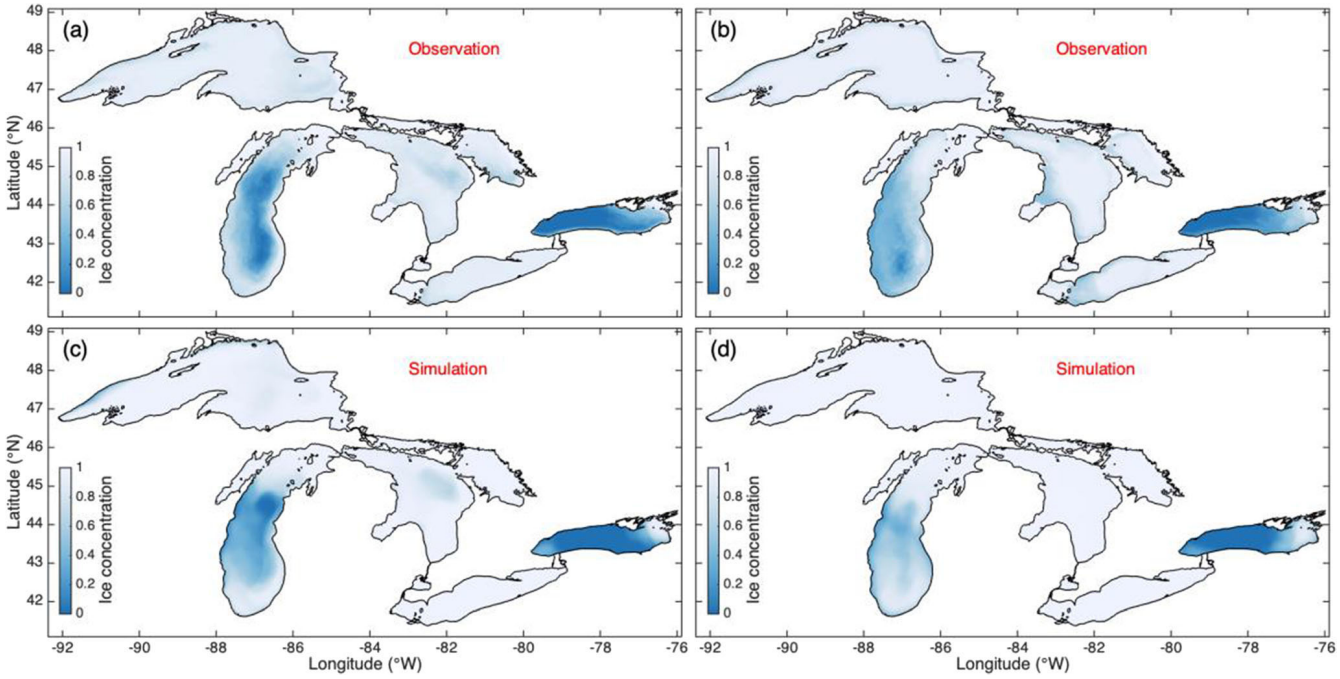
How to read this figure

NARR wind time series are compared with buoy winds, with mean values shown for each record. The tendency toward weaker modeled forcing helps explain background wave errors and motivates controlling forcing quality when assessing a new ice parameterization or surrogate.

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FIGURE ATLAS | 05 / 07

Figure 5. Observed and modeled ice concentration



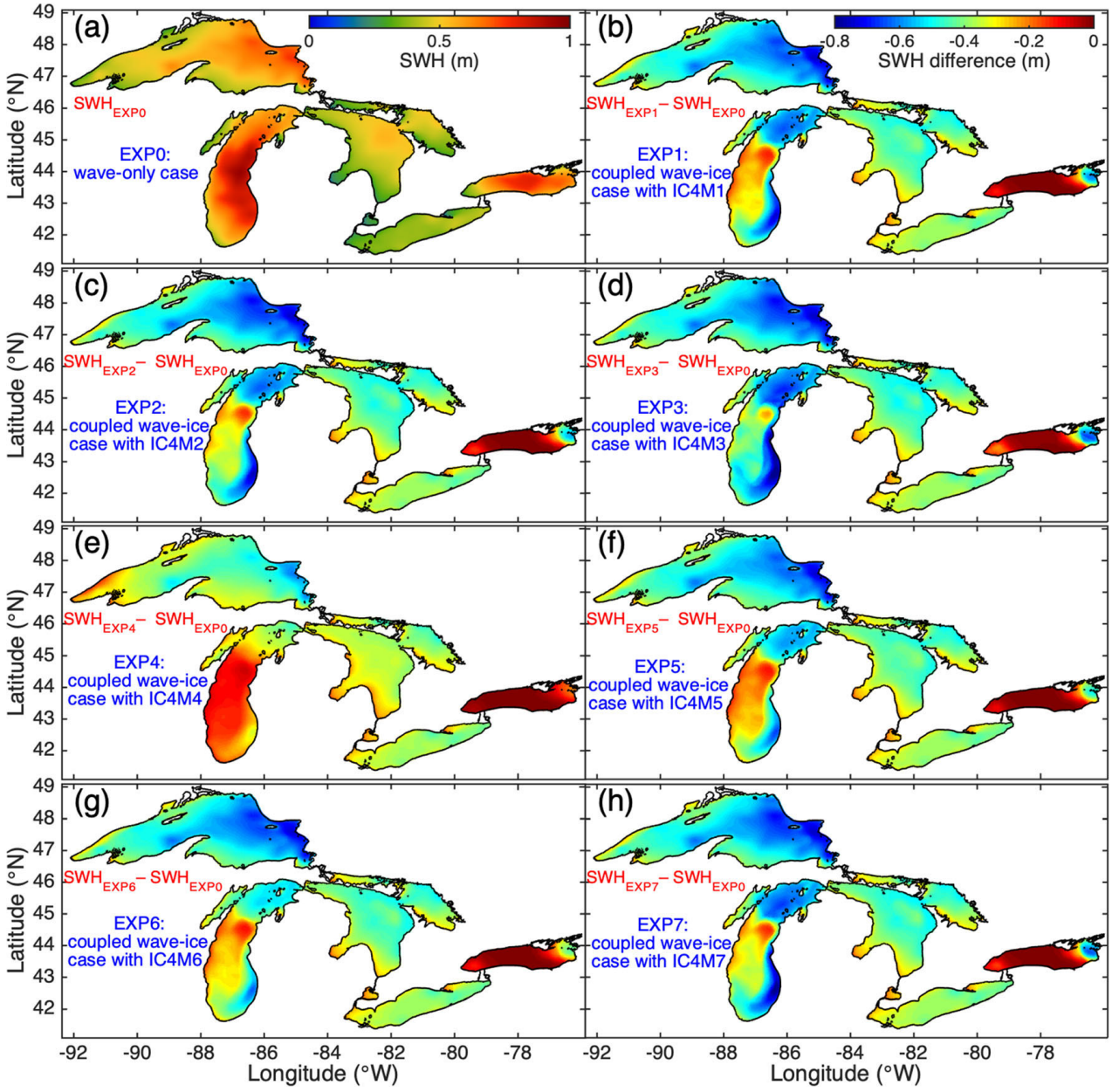
How to read this figure

NIC maps and modeled monthly ice concentration for February and March 2014 show the broad spatial distribution and seasonal progression of ice. Because concentration controls blocking and wave damping, an incorrect ice field can overwhelm differences among attenuation formulas.

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FIGURE ATLAS | 06 / 07

Figure 6. Basin-wide sensitivity to the seven IC4 schemes



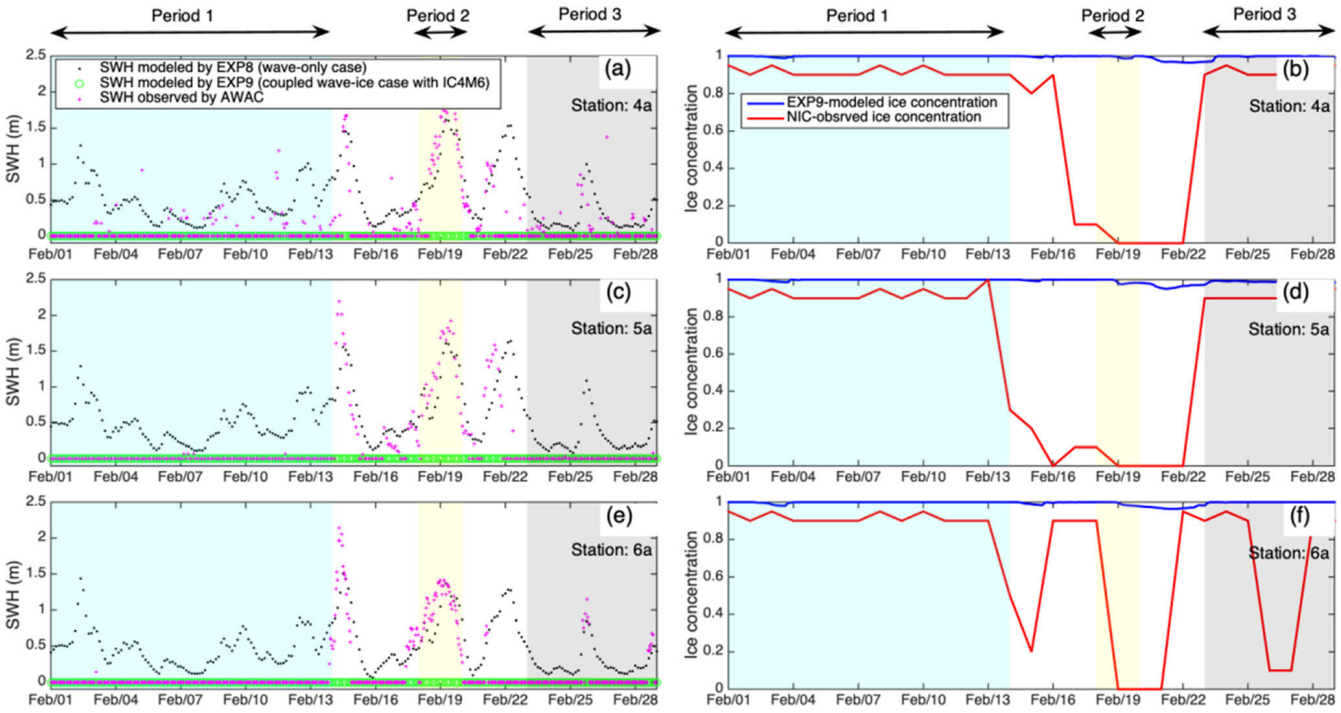
How to read this figure

The wave-only reference map is compared with wave-height differences from the seven ice-aware simulations. All produce stronger suppression in ice-rich lakes, while the magnitude varies by scheme. These are model-based sensitivity estimates for 2014, rather than seven independently validated winter forecasts.

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FIGURE ATLAS | 07 / 07

Figure 7. Winter validation and the rapid-retreat failure



How to read this figure

AWAC wave heights and NIC ice concentrations expose three distinct intervals in February 2011. The ice-aware model improves high-ice conditions but misses the large waves associated with the short retreat event. Read the ice and wave panels together: they identify forcing and coupling errors that aggregate wave metrics alone would obscure.

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